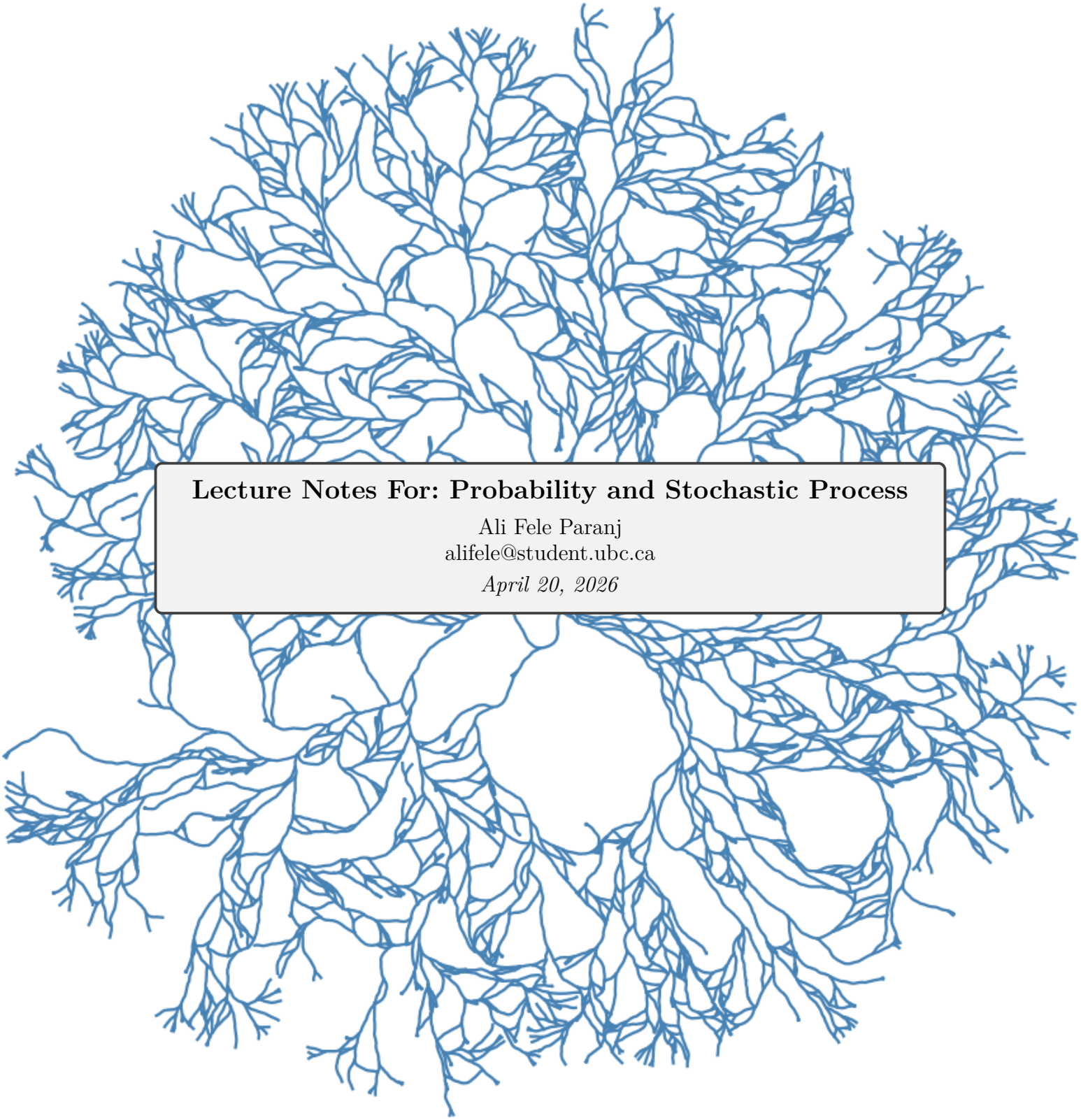

Lecture Notes For: Probability and Stochastic Process

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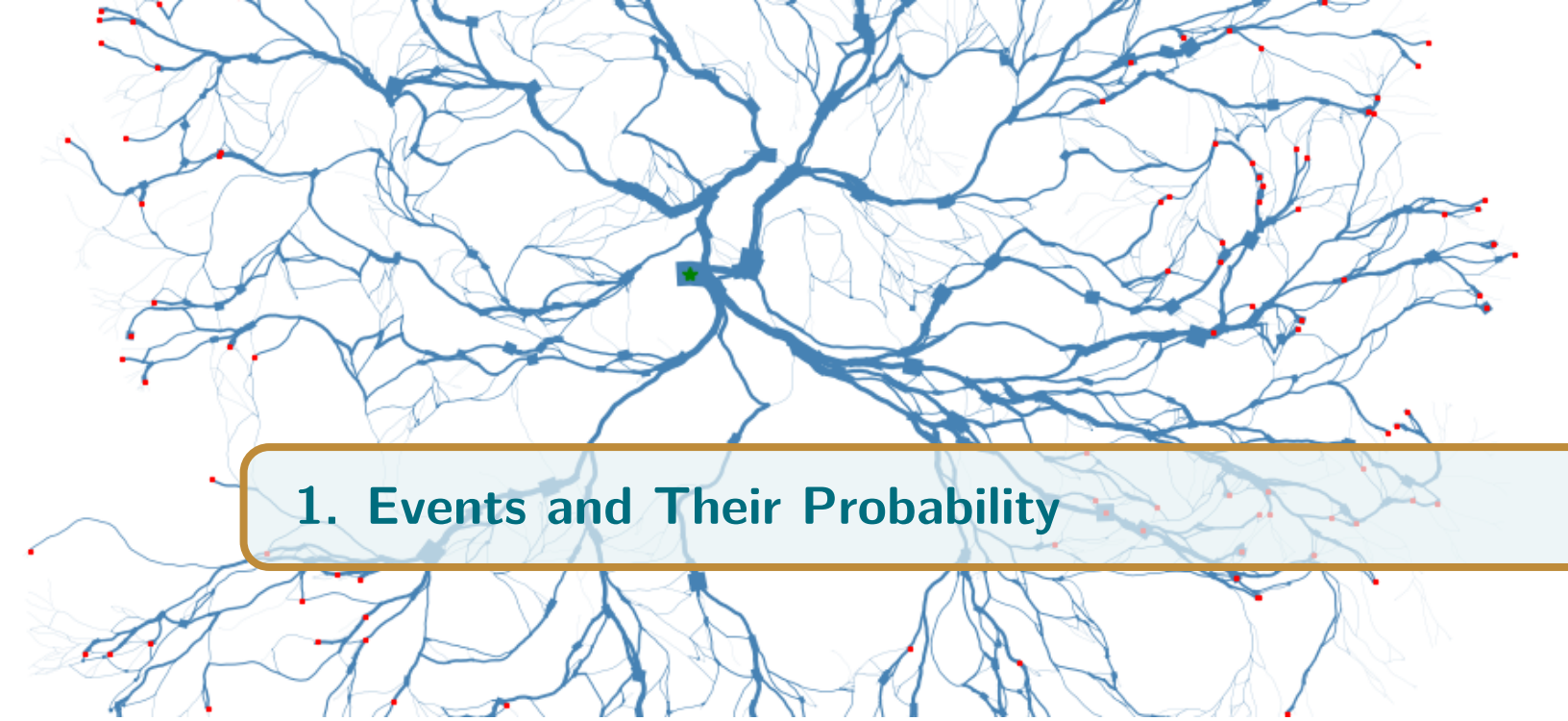


1. Events and Their Probability

1.1 Events as Sets

■ **Problem 1.1** Let $\{A_i : i \in I\}$ be a collection of sets. Prove "De Morgan's Laws"'

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \qquad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

Solution Let $x \in (\cup_i A_i)^c$. Then $x \notin \cup_i A_i$. So $\forall i \in I$ we have $x \notin A_i$, or equivalently $x \in A_i^c$. Using the definition of intersection $x \in \cap_i A_i^c$. For the second identity,

$$\begin{aligned}
&x \in \left(\bigcap_i A_i\right)^c && \Leftrightarrow \\
&x \notin \bigcap_i A_i && \Leftrightarrow \\
&\exists i \in I \text{ s.t. } x \notin A_i && \Leftrightarrow \\
&\exists i \in I \text{ s.t. } x \in A_i^c && \Leftrightarrow \\
&x \in \bigcup_i A_i^c.
\end{aligned}$$

■ **Problem 1.2** Let A and B belong to some σ-filed $\mathcal{F}$. Show that $\mathcal{F}$ contains the sets $A \cap B$, $A \backslash B$ and $A \Delta B$.

Solution Since $A, B \in \mathcal{F}$, $A^c, B^c \in \mathcal{F}$ as well. So

$$A \cap B = (A^c \cup B^c)^c \in \mathcal{F},$$

since $A^c, B^c \in \mathcal{F}$ and $\mathcal{F}$ is closed under union and complement. Furthermore

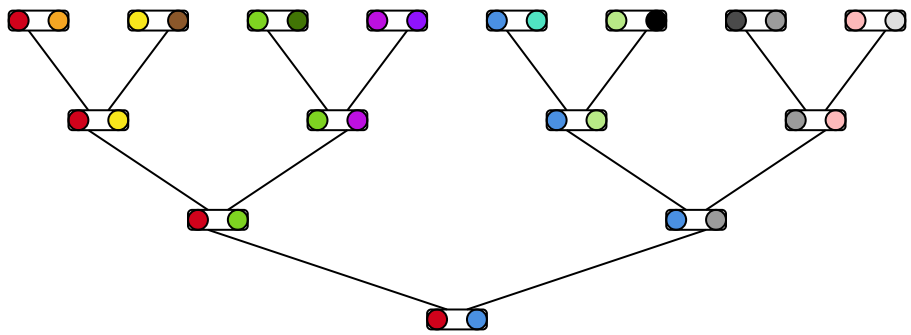
$$A \backslash B = A \cap B^c \in \mathcal{F}.$$

And

$$A \Delta B = (A \cup B) \backslash (A \cap B) \in \mathcal{F}.$$

■ **Problem 1.3** A conventional knock-out tournament begins with 2^n competitors and has n rounds. There are no play-offs of the positions $2, 3, \cdots, 2^n - 1$, and the initial table of the draws is specified. Gen a concise description of the sample space of all possible outcomes.

Solution Consider the following draw table for the tournament. Without loss of generality we can assume that the team on the left will win each game (as depicted in the following draw table). Then each placement of the teams in the first row represents a valid outcome of the tournament. So for 2^n players, there will be $2^n!$ possible outcomes for the tournament.



■ **Problem 1.4** Let $\mathcal{F}$ be a σ-field of subset of ω and suppose that $B \in \mathcal{F}$. Show that $\mathcal{G} = \{A \cap B : A \in \mathcal{F}\}$ is a σ-field of subsets of B.

Solution Any element of $\mathcal{G}$ is of the form $A \cap B$ for some $A \in \mathcal{F}$, thus $A \cap B \subset B$. It remains to show that G is a σ-field.

(a) Contains B and $\varnothing$: Because $\Omega \in \mathcal{F}$, so $\mathcal{G} \ni \Omega \cap B = B$. Furthermore, $\varnothing \in \mathcal{F}$. So $\mathcal{G} \ni \varnothing \cap B = \varnothing$.

(b) Is closed under countable union: Let $(A_i \cap B) \in \mathcal{G}$. Then

$$\bigcup_i (A_i \cap B) = (\bigcup_i A_i) \cap B \in \mathcal{G}.$$

(c) Is closed under complement with respect to B: Let $A \cap B \in \mathcal{G}$. The complement of $A \cap B$ with respect to B is

$$B \backslash (A \cap B) = B \cap (A \cap B) = B \cap (A^c \cup B) = A^c \cap B \in \mathcal{G}.$$

■ **Problem 1.5** Which of the following are identically true? For those that are not, say when they are true.

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

(b) $A \cap (B \cap C) = (A \cap B) \cap C$.

(c) $(A \cup B) \cap C = A \cup (B \cap C)$.

(d) $A \backslash (B \cap C) = (A \backslash B) \cup (A \backslash C)$.

Solution (a) True.

(b) True.

(c) Not true. Only holds when $C \subset B$.

(d) True.

1.2 Probability

■ **Problem 1.6** Let A and B be events with probabilities. Show that

$$\frac{1}{2} \leq \mathbb{P}(A \cap B) \leq \frac{1}{3}.$$